



CURRICULUM OVERVIEW

Little Coders Path

Build 3 Real Projects

A gentle, hands-on introduction to computing for young learners. Students move from colorful visual blocks to real Python code and even a first taste of AI — building genuine, shareable projects the whole way, at a pace made for ages 6–11.

3

Courses

24

Lessons Total

3

Real Projects

6–11

Age Range

LEARNING PROGRESSION

10**Scratch Adventures**

Animated stories, games, and interactive projects — no typing needed.

**11****Python for Young Builders**

A gentle bridge from visual blocks to real, typed Python programs.

**12****AI for Kids: Smart Projects**

Train real AI models and connect them to fun, interactive apps.

Learn by playing, not by lecture. Every lesson turns a new idea into a game, story, or project your child actually wants to finish.

A portfolio, not just a certificate. Students finish with three real projects — animations, programs, and AI apps — they're proud to share.

COURSE 10 · LESSONS 1–8

Scratch Adventures

Create animated stories, games, and interactive projects visually

Scratch (MIT)

Scratch Editor

Scratch Community

OVERVIEW

The perfect first step for young learners. Scratch uses visual blocks instead of text, so your child focuses on logic and creativity, not syntax. By the end, they have real animated projects to show off.

PORTFOLIO PROJECTS

- **Animated animal story**
- **Interactive quiz game**
- **Creative music maker**

WHY IT MATTERS

Every programmer starts with the same big ideas: sequence, events, loops, and decisions. Scratch teaches these ideas visually, so young learners build real logical thinking before they ever type a line of code.

LESSON BREAKDOWN

- 1 What is coding and why does it matter**
- 2 Scratch setup and first sprite**
- 3 Motion, looks, and sounds blocks**
- 4 Events and broadcasting**
- 5 Loops and repetition**
- 6 Conditionals and decisions**
- 7 Building an interactive story**
- 8 Final project: our own game**

COURSE 11 · LESSONS 1–8

Python for Young Builders

Write first real Python programs (the same language used by NASA and Google)

Python 3

Thonny IDE

Repl.it

OVERVIEW

The bridge from visual to text-based coding. Your child writes their first Python programs — variables, loops, and their very own mini projects. Taught slowly, with patience, and lots of fun.

PORTFOLIO PROJECTS

- Personal quiz app
- Simple calculator
- Word guessing game

WHY IT MATTERS

Moving from visual blocks to typed code is a big milestone. This course makes that jump gently, showing students that the logic they already know from Scratch works exactly the same way in real, professional code.

LESSON BREAKDOWN

1 From Scratch blocks to Python text

2 Variables and user input

3 Printing and string formatting

4 If/else decisions

5 While loops and for loops

6 Lists and simple data

7 Building a mini quiz app

8 Final project: our own mini game

COURSE 12 · LESSONS 1–8

AI for Kids: Smart Projects

Train your own AI models and build apps that see, listen, and respond

Scratch

Teachable Machine

ML4Kids

Google AI Tools

OVERVIEW

Your child discovers how AI actually works, not by reading about it, but by building with it. Using Google's Teachable Machine and ML4Kids, they train machine learning models that recognize images and sounds, then connect them to Scratch to build interactive AI-powered projects any parent will be amazed by.

PORTFOLIO PROJECTS

- AI rock-paper-scissors game
- Voice-activated Scratch story
- Emotion-recognition virtual pet

WHY IT MATTERS

AI already shapes the apps and devices kids use every day. This course demystifies it early: students see, hands-on, that AI is trained by people using real examples — not magic — and that they can build with it themselves.

LESSON BREAKDOWN

- 1 What is AI and how do machines actually learn
- 2 Training your first image recognition model
- 3 Connecting an AI model to Scratch with ML4Kids
- 4 Building an AI rock-paper-scissors game
- 5 Voice commands: training a sound recognizer
- 6 Creating a voice-activated story in Scratch
- 7 Facial expression recognition and virtual pets
- 8 Showcase: presenting your AI project to the world

The Complete Journey

Across 3 courses and 24 lessons, students progress from their first visual coding blocks to real Python programs and hands-on AI projects — walking away with three real, shareable creations for their portfolio.

COURSE 10

Scratch Adventures

- Sprites, motion, and sounds
- Events and broadcasting
- Loops and conditionals
- Building an interactive game

COURSE 11

Python for Young Builders

- Variables and user input
- If/else decisions
- While loops and for loops
- Building a mini quiz app & game

COURSE 12

AI for Kids

- How machines learn from examples
- Training image & sound models
- Connecting AI to Scratch
- Showcasing a real AI project

WHAT STUDENTS BUILD

1

Scratch Portfolio

Animated animal story · Interactive quiz game · Creative music maker

2

Python Portfolio

Personal quiz app · Simple calculator · Word guessing game

3

AI Portfolio

AI rock-paper-scissors game · Voice-activated Scratch story · Emotion-recognition virtual pet

BY THE END OF THE PATH, STUDENTS CAN



Build an animated story or game in Scratch using loops, events, and conditionals.



Write and run their own real Python programs, including variables, loops, and decisions.



Train a simple AI model and connect it to a fun, interactive project of their own.

Ready to start building?

All lessons and projects are taught live by an expert Lumexa instructor.